

Kinect v2 tracked Body Joint Smoothing for Kinematic Analysis in Musculoskeletal Disorders

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Abstract—Body joint monitoring is essential for disorder diagnosis and assessment of treatment effectiveness. Microsoft Kinect v2 is a low-cost and markerless human motion-tracking RGB-D sensor that provides spatial locations of tracked skeletal joints in the form of 3D coordinates. Sometimes, body tracking of kinect v2 produces erratic 3D coordinates, which affects the real-time tracking performance of the sensor. A careful study of the literature suggests that skeletal tracking of kinect v2 needs further exploration. This work proposes a filter combined with the concept of body kinematics to remove noise and enhances the quality of 3D coordinates in body frame data. Also, it generates “Motion Signature” of the tracked joint, which shows movement pattern for kinematic analysis, and helpful in joint monitoring of Musculoskeletal Disorders (MSD). The clinically relevant anatomical movement was executed, to evaluate the performance of the proposed filter. We compared Range of Motion (RoM) values obtained from the proposed filter with the gold standard goniometry. Results indicate that RoM values from the proposed filter are in high correlation with the goniometry with an Intraclass Correlation Coefficient values ranging between 0.95 to 0.98 authenticating that it improves the skeletal joint tracking of kinect v2.

Clinical relevance—The proposed filter improves the skeletal joint tracking of kinect v2. It can be used for body joint monitoring, identify and evaluate individuals with musculoskeletal disorders, and treatment effectiveness. The proposed filter also finds applications in sports biomechanics for player movement monitoring.

I. INTRODUCTION

Musculoskeletal (MSK) system provides strength and assistance to the human body. Musculoskeletal Disorder (MSD) is an injury that affects the MSK system. It is a non-communicable degenerative disorder with global prevalence between 28% to 96%, and the second leading cause of years lived with disability followed by neuropsychiatric disorders [1]. Deformity and swelling in joints lead to stiffness, and at a later stage, cause pain and reduced functionality. It accounts for 61.4% of globally disability-adjusted life years (DALYs) in 2016 and primarily affects the efficiency and productivity of the population [2]. Low Back Pain, neck pain, arthritis, adhesive capsulitis are primary chronic conditions that reduce the quality of life of an individual. MSK joints provide mobility and functionality to execute daily life tasks.

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Its monitoring and diagnosis are essential steps to maintain sound MSK health. The joint monitoring takes place in physical rehabilitation and occupational activities, and to measure treatment effectiveness after joint surgery. Joint Range of Motion (RoM) is a clinical parameter that informs the amount of movement a body joint can achieve with its maximum potential, and its kinematic analysis deduces clinical information related to movement patterns [3].

Universal Goniometer (UG) is a standard tool used by clinicians to measure RoM. Although it shows excellent intra-rater (Degree of agreement in repeatedly measured parameter values from single rater) and inter-rater (Degree of agreement in repeated measured parameter values from various assessors) reliability, there are some limitations in its use. The desired movement for RoM measurement is to be fixed at the same position for some duration, requires clinician intervention, and accurate positioning of it over body joint. Misalignment of UG with joint location can lead to measurement errors. As a result, this leads to the development of viable alternative tools for RoM assessment.

Technological development in motion-sensing devices enables us to continuously monitor and track the movements, which helps to extract clinical features for diagnosis and treatment. There are mainly two types of motion sensing devices, which continuously monitor joint movement data collection, namely wearable and optical. Examples of wearable sensors are accelerometer, gyroscope *etc.* [4]. The optical sensor has two types, *viz.* marker-based and marker-less systems. The marker-based system can be further classified into an active and passive marker system, which utilizes a light-emitting and retro-reflective marker, respectively. Examples of marker-based motion-sensing systems are Optotrak, Vicon *etc.* [4]. Such sophisticated systems are efficient and provide quantitative clinical information about movement, but not user friendly because it demands bio-markers to adhere to the subject's joint location, involves high installation cost, and an expert for system operation. Such requirements restrict its use in some clinical, sports, and home environments.

Microsoft Kinect v2 is a low-cost, easy to set up, and vision-based marker-less motion sensor, based on Natural User Interface (NUI) [5]. It facilitates all the limitations of both wearable and sophisticated optical motion-sensing devices. Its output is different raw data streams of the tracked subject at a rate of 30 frames per second, namely body, color, depth, and infrared. The body data streams provide hypothesized body joint locations for each frame, which are 3D spatial location and corresponding coordinate values based on inferred body parts from depth image space points

[5]. The second version (v2) shows improved color and depth resolution and 25 joint tracking capabilities. It has proven advantageous in terms of portability and usability compared to other sophisticated motion capture systems. Body tracking provides frame-by-frame skeletal joint positions with acceptable accuracy, but not necessarily precise, and causes erratic 3D joint coordinates. The various factors, which affect the sensor's precision (repeatability in inference of skeletal joint 3D position) are, namely occlusion by body part or joint, incorrect pose, cases where the skeletal joint position is outside from the sensor's field of view, the distance of a human body from sensor *etc* [6]. Such scenarios cause the incorrect skeletal joint inference, which leads to jitters, and results in unusual tracking and shifting behavior of skeletal joint position. Several solutions are proposed in this direction. Schmitz *et al.* [10] has validated the accuracy and repeatability of kinect to measure knee joint angles at the different poses. Here authors used the jig to model the knee joint. Adjeisah *et al.* [7] have proposed holt exponential smoothing filter [8] to smooth jitters present in tracked skeletal joints for gesture recognition in NUI based applications. Performance analysis of kinect v2 in the context of the clinical use was presented by Napoli *et al.* [9], but reported lower skeletal joint angle accuracy with respect to Qualisys, which is a multiple cameras based motion tracking system. The careful literature review indicates that skeletal joint tracking of kinect v2 for anatomical movements in clinical and sports settings needs further exploration.

This work proposes a method to generate "Motion Signature" of skeletal joint, constructed by its RoM data, that shows motion pattern, and helpful in kinematic analysis. For the above purpose, the proposed filter has a correction and smoothing steps to update body skeletal joint 3D coordinates of kinect v2. It temporarily stores the previous 3D coordinate values of the tracked skeletal joint, and smooths the data. Secondly, RoM values of tracked joints are stored and further utilized to generate motion signature. The skeletal joint correction and smoothing procedure have been applied separately on X, Y, and Z coordinates to improve the body tracking performance. Motion Signature provides details of movement patterns and can assist clinicians to evaluate the body joint performance.

The remaining paper is organized as follows; Section II explains methodology, and section III shows the evaluation and results from proposed filter. Section IV includes discussion, and section V concludes the paper.

II. METHODS

A. Anatomical Movements and Data Acquisition

Body joints move in Human Anatomical Planes, namely Frontal or Coronal (X-Y plane), Sagittal (Y-Z plane), and Transverse or Horizontal (X-Z plane), as shown in Fig. 1. The body tracking of kinect v2 provides noisy skeletal data that affect sensor's precision and tracking performance [11]. Therefore, filtering of skeleton 3D coordinates is required. We have targeted shoulder and elbow joints position data

for filtering, and anatomical movements, namely lateral abduction in the frontal plane, anterior flexion and posterior extension in the sagittal plane, and medial-lateral external rotation in the transverse plane to evaluate proposed filter's performance.

The user interface (UI) for skeletal joint tracking was developed in C# with Visual Studio Community IDE (Microsoft Corporation, USA) and Software Development Kit 2.0 (SDK) of kinect v2. It displays (1) acquired color and body frame, and (2) skeletal joint positions of the tracked body. We have added a feature to display and store real-time RoM values of body skeletal joint to generate a motion signature for kinematic analysis.

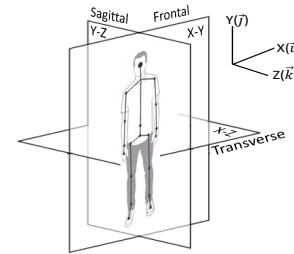


Fig. 1: Human Anatomical Plane for movement

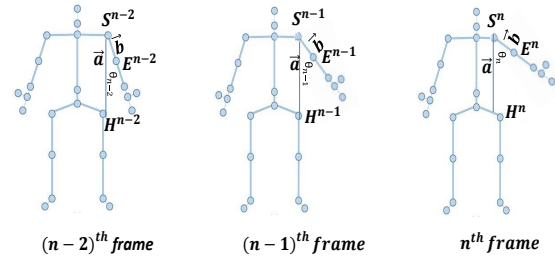


Fig. 2: Skeletal joints used in filtering

B. Proposed Method : Skeletal Joint Smoothing

It is reported in [12] that kinect v2 classify human body-parts, frame-by-frame, from depth images, and provides 25 skeletal joints with 3D spatial positions. For the proposed work of shoulder joint motion analysis, shoulder, hip, and elbow joints are denoted by S_n , E_n , and H_n at n^{th} frame, as shown in Fig. 2. Let the 3D coordinates of S^n be given as (x_n^s, y_n^s, z_n^s) . Similarly, 3D coordinates of elbow and hip are given as (x_n^e, y_n^e, z_n^e) , (x_n^h, y_n^h, z_n^h) , respectively. Also, let there be a vector $\vec{a}$, from S^n to H^n , and $\vec{b}$, from S^n to E^n , as shown in Fig. 2, given below:

$$\vec{a} = (x_n^s - x_n^h)\hat{i} + (y_n^s - y_n^h)\hat{j} + (z_n^s - z_n^h)\hat{k} \quad (1)$$

$$\vec{b} = (x_n^s - x_n^e)\hat{i} + (y_n^s - y_n^e)\hat{j} + (z_n^s - z_n^e)\hat{k} \quad (2)$$

Angle θ for RoM measurement between the two vectors $\vec{a}$ and $\vec{b}$, as shown in Fig. 2, can be found as follows:

$$angle(\theta) = \arccos \left(\frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\| \|\vec{b}\|} \right) \quad (3)$$

The 3D coordinates of shoulder (S_n), hip (H_n), and elbow (E_n) joints in the n^{th} frame can be denoted collectively as C_n . Noise in skeletal joint tracking causes erratic 3D coordinates, and to address this problem, the following filter is proposed.

Firstly, we used body kinematics in the proposed filter, to estimate shoulder-elbow bone length at the initial step, given in (4).

$$\|b\| = \sqrt{(x_n^s - x_n^e)^2 + (y_n^s - y_n^e)^2 + (z_n^s - z_n^e)^2} \quad (4)$$

The difference in $\|b\|$ in subsequent frames evoke the proposed filter for 3D joint coordinates correction. The pairwise difference of 3D coordinates corresponding to shoulder, elbow, and hip joints, were obtained by recent three frames, as given in (5) and (6), respectively.

$$D_1 = C_n - C_{n-1} \quad (5)$$

$$D_2 = C_{n-1} - C_{n-2} \quad (6)$$

Secondly, we proposed a hypothesis that for clinical movement monitoring, coordinates from n^{th} frame to $(n+1)^{th}$ frame do not differ substantially. Hence, an average of the estimated quantum of change in coordinates in the previous frames, given in (5) and (6), is used to correct 3D coordinate point in the $(n+1)^{th}$ frame. Hence, the updated 3D coordinates of skeletal joint C_n at $(n+1)^{th}$ frame can be obtained from (7).

$$\hat{C}_{n+1} = C_n + \frac{D_1 + D_2}{2} \quad (7)$$

Algorithm 1: Skeletal Joint Smoothing

Input: Raw 3D coordinate values

Output: Updated 3D coordinate values

```

1 while body frame != NULL do
2   for frame = 1,2,...,nth // body tracking frames
3   Acquire BodyFrame // body frame acquisition
4   for each skeletal joint in body do
5     acquire 3D coordinates of shoulder and elbow
      joints;
6     Cn ← Assign 3D coordinate values
7     calculate ||b|| for each frame;
8     for each Co-ordinate in Cnje and Cnjs do
9       Calculate ||b|| based on (4)
          //  $\vec{b}$  = Shoulder Elbow Vector
10      if ||bn+1|| ≠ ||bn|| then
11        Update shoulder and elbow joint 3D
          coordinates based on (7);
12      return  $\hat{C}_{n+1}$ ;
13      RoM calculation  $\theta$  based in (3);
14      frame ++;
```

TABLE I: Intra Class Coefficient Values

ICC values of RoM measurement between goniometry and proposed filter				
Motion	Goniometry-Proposed Filter ICC	Low 95% CI	High 95% CI	Mean Diff. (Degree)
Abduction	0.98	0.87	0.99	3.8
Flexion	0.96	0.77	0.99	5.6
Extension	0.95	0.73	0.99	6.1
External Rotation	0.98	0.88	0.99	4.5

III. EVALUATION AND RESULTS

The shoulder movements, namely abduction, flexion and extension, and lateral-medial external rotation, were executed to evaluate the effectiveness of the proposed filtering method. For each type of anatomical movement, five different movements were executed with ascending RoM values. RoM values were obtained from UG (with two different observers), and the proposed filter. Intraclass Correlation (ICC) [13], which is a modification of the Pearson correlation coefficient, was used to find agreement between the RoM values obtained from both the methods. Two-way random effects, absolute agreement, and multiple measurement model ICC (2,k) [13], was chosen in this study. The MedClac software (Ostend, Belgium) was used, and confidence interval (CI) was set to 95% for statistical analysis. Table I shows that ICC values are in the range between 0.95 to 0.98. and mean differences of RoM measurements are in the range between 3.8 to 6.1 degree, which shows high agreement for RoM values between the proposed filter and gold standard goniometry. Fig. 3 shows the Box and Whisker Plot that summarizes and compare RoM values obtained from goniometry and proposed filter. Fig. 4 shows shoulder-elbow bone length variation before and after the smoothing of skeletal joints in external rotation. Further, Fig. 5 shows generated motion signature of the shoulder joint in abduction, which can be helpful to extract clinical features for kinematic analysis, namely execution time, time-range of normal and abnormal movement of skeletal joint, and maximum RoM achieved. Fig. 6 shows 3D scatter plot of raw and smoothed 3D coordinates of the right elbow joint during the abduction.

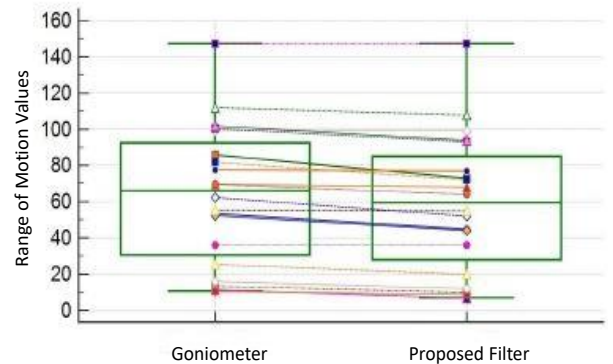


Fig. 3: Box and Whisker Plot

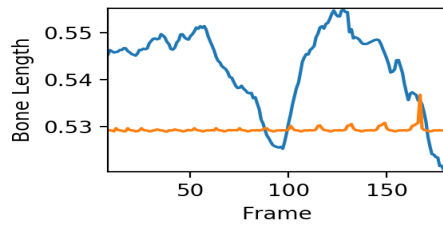


Fig. 4: *S-E* bone length variation in external rotation

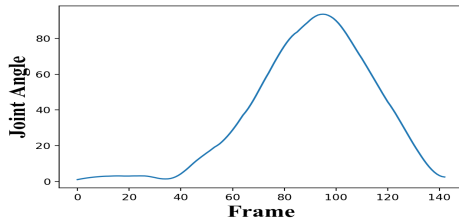


Fig. 5: Shoulder joint motion signature in abduction

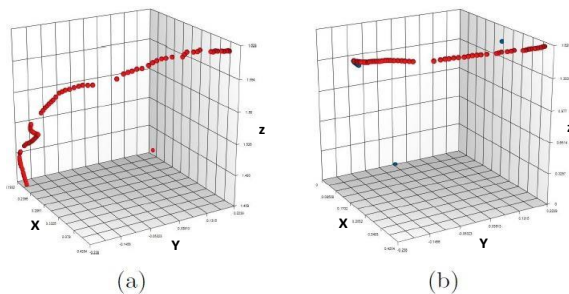


Fig. 6: 3D Scatter Plot of (a) Raw and (b) Smooth 3D coordinates of elbow joint in shoulder abduction

IV. DISCUSSIONS

The proposed filter removes noise and smooth the skeletal joints of kinect v2, and extends its avenues in sports and clinical environments. The objective of this work is to propose a novel filter for noise reduction and provide smoothed skeletal 3D coordinates of tracked skeletal joints for movement tracking and kinematic analysis. RoM values obtained from the proposed filter are in high correlation with gold standard goniometry. It also generates motion signature of skeletal joint movement, which can be used to track MSK health, extract clinical features for movement monitoring and analysis. Also, it finds applications in the preliminary diagnosis of patients suffering from MSD. The ICC values are in the range between 0.95 to 0.98, which show excellent reliability of proposed filter's result as compared with [14] and [15]. Hence, results show the proposed filter's potential to improve the tracking performance of kinect v2.

V. CONCLUSIONS

This work proposes a filter to remove noise and smooths 3D coordinates of tracked skeletal joints positions to generate motion signature for kinematic analysis of body joint movement. Table I, and Fig. 3, 4, 5, and 6 show the results of

proposed filter. The generated signatures can assist clinicians in extracting clinical features for diagnosis and monitoring of body joint movements. The limitation of the proposed filter is its inability to provide promising results in smoothing of lower limb skeletal joints. However, with modification in proposed filter, its application could be extended to improve the tracking performance of lower limb joints.

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